

Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Performance vs Fan Attendance in the Premier League

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2024-2025 Season | SIS No. 1 | Data Collection & Preparation

1. Introduction and Objective

This study examines whether higher average home attendance is associated with better results in the English Premier League during the 2024-2025 season. Club performance is measured by total points, number of wins, goal difference, and final league position. The objective is to collect two independent real-world datasets, prepare and combine them, and evaluate their statistical relationships.

2. Data Sources and Collection

Football performance data were retrieved from the Football-data.org REST API using the Premier League standings endpoint (/v4/competitions/PL/standings?season=2024). The response provided standings and match statistics for 20 clubs, including 38 games played per club. Attendance data were collected from the 2024-25 Premier League Wikipedia season page by using requests and BeautifulSoup to parse the HTML attendance table. The collection script checked the site robots.txt rules before making the page request. The scraped table contained the average home attendance and 19 home games for each club.

3. Cleaning, Transformation and Merging

Using pandas, both datasets were checked for missing values, duplicated records and appropriate numeric data types. Club names were normalized by converting them to lowercase, removing common prefixes/suffixes such as FC or AFC, and harmonizing punctuation and ampersands. A one-to-one outer merge on the cleaned club names matched all 20 clubs with no unmatched rows. The final dataset contained 20 rows and 12 columns, with no missing values.

4. Exploratory Analysis

Descriptive statistics were calculated for attendance and football performance. Pearson correlation coefficients were computed to measure the direction and strength of linear relationships between average home attendance and the four performance measures. Five labeled visualizations were created in Matplotlib and Seaborn.

5. Key Descriptive Statistics

Measure	Result
Average attendance across clubs	40,400.7 fans
Median attendance across clubs	35,327.5 fans
Highest average attendance	Manchester United - 73,747
Lowest average attendance	AFC Bournemouth - 11,214
Average league points	52.35 points

Note: attendance refers to the average number of spectators at a club's home league matches.

6. Correlation Results

Pair of variables	Pearson r	Interpretation
Attendance vs. league points	+0.206	Weak positive
Attendance vs. wins	+0.228	Weak positive
Attendance vs. league position	-0.197	Very weak negative
Attendance vs. goal difference	+0.269	Weak positive

For league position, a smaller number indicates a better finish; therefore, a negative coefficient points toward a slightly better ranking.

7. Results and Discussion

The relationships are weak overall. Manchester United had the highest average attendance (73,747) but finished 15th with 42 points. Liverpool attracted 60,330 spectators per home game on average and won the league with 84 points. In contrast, Bournemouth had the lowest average attendance (11,214) yet finished 9th with 56 points. These examples show that a large home crowd does not necessarily translate into a stronger league performance.

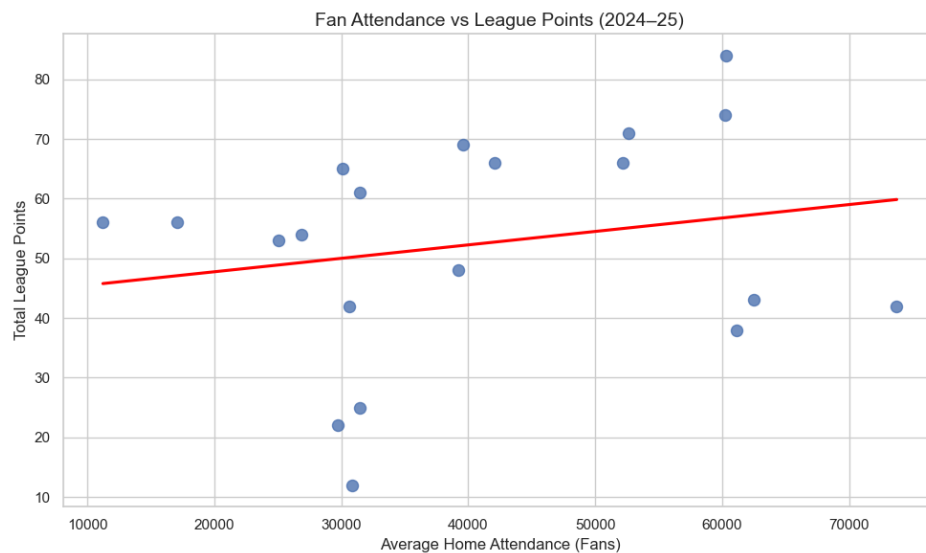


Figure 1. Average home attendance vs. league points (source: project Notebook).

8. Limitations and Conclusion

This analysis covers one season and uses club-level average attendance rather than match-by-match crowds. Stadium capacity, team quality, spending, and other factors may affect both attendance and results; they were not controlled for here. Thus, the observed correlations cannot establish causation. The project successfully demonstrates API collection, respectful HTML scraping, data cleaning, one-to-one merging, basic EDA and visualization. The main conclusion is that attendance shows only a weak positive association with points ($r = 0.206$), so attendance alone is not a reliable indicator of Premier League performance.

References

Football-data.org. API documentation and Premier League standings: <https://www.football-data.org/documentation/quickstart>
Wikipedia contributors. "2024-25 Premier League": https://en.wikipedia.org/wiki/2024%E2%80%9325_Premier_League